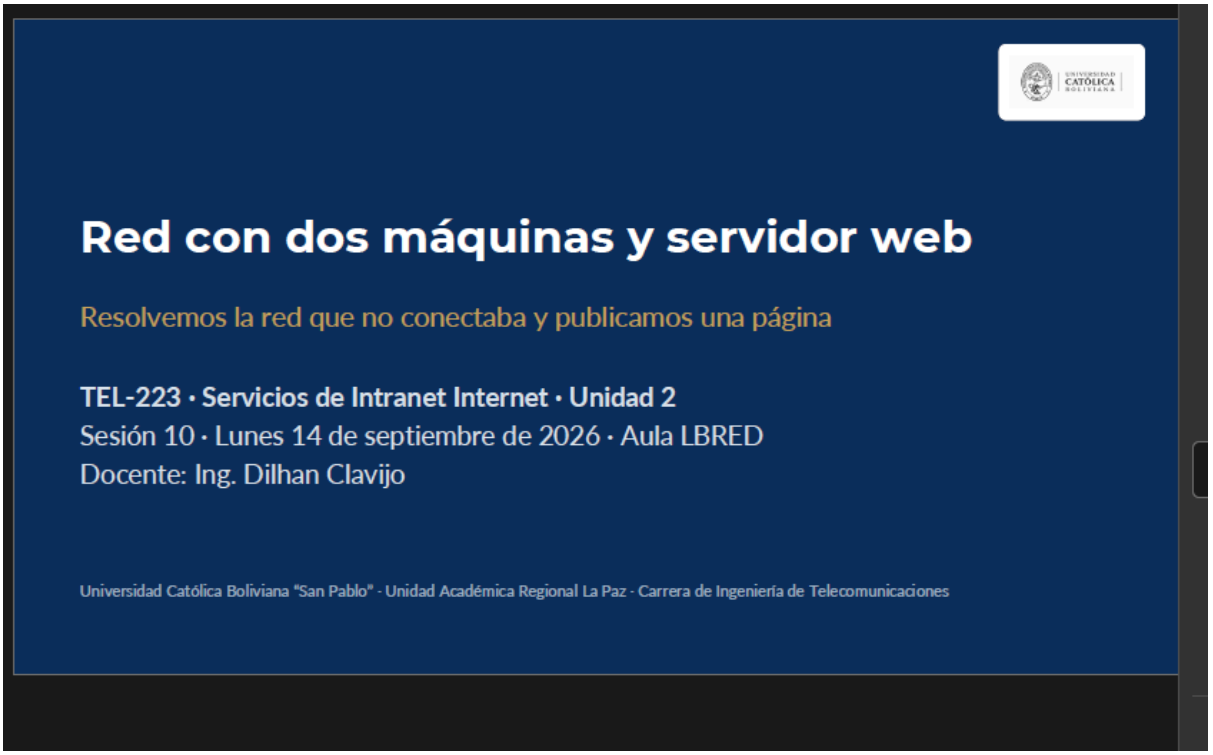
UNIVERSIDAD CATOLICA BOLIVIANA ”SAN PABLO”

UNIDAD ACADÉMICA REGIONAL LA PAZ

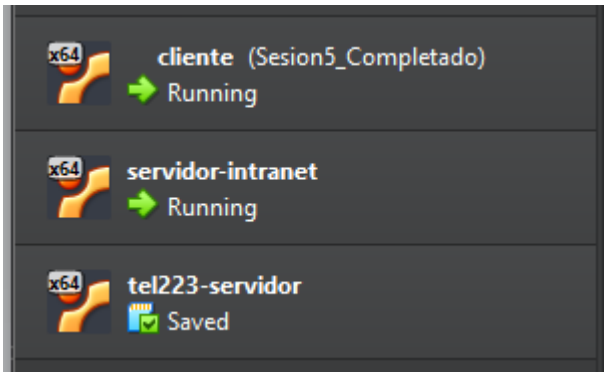
INGENIERÍA DE TELECOMUNICACIONES

Laboratorio ·

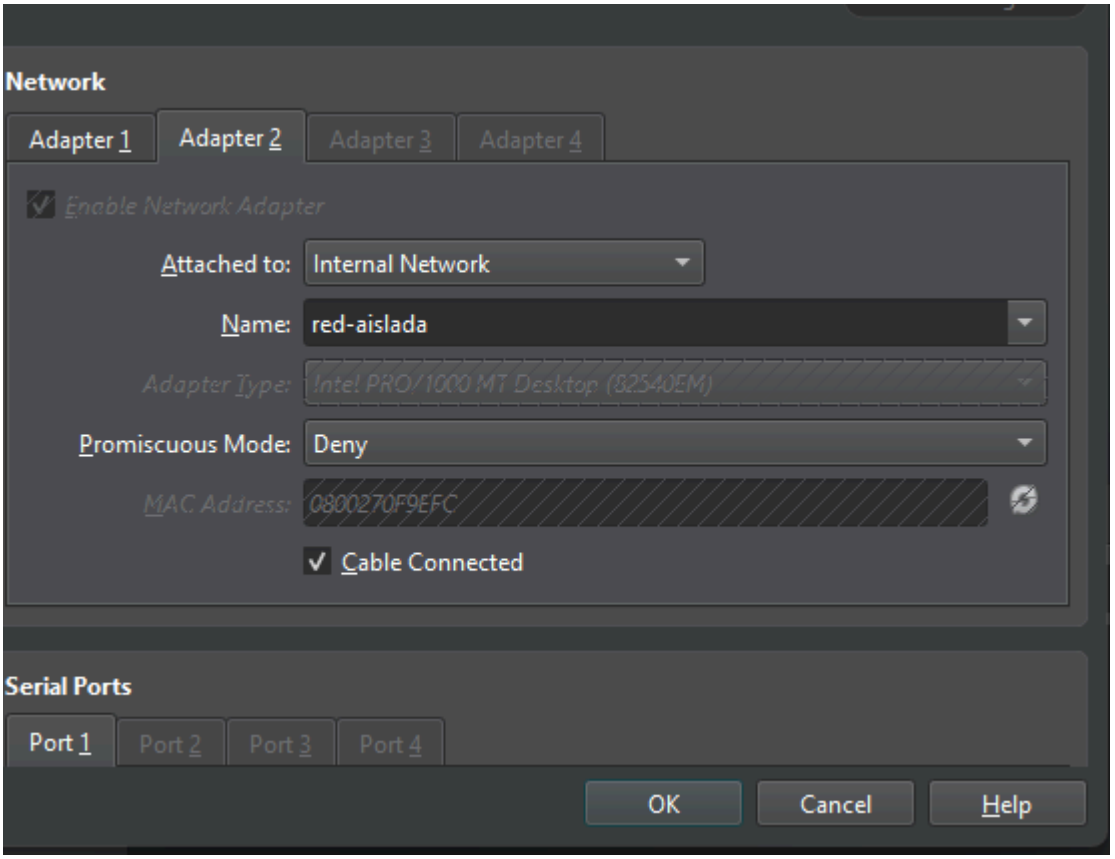
NOMBRE **Carlos** **Andres** **Fernandez** **Guzman**

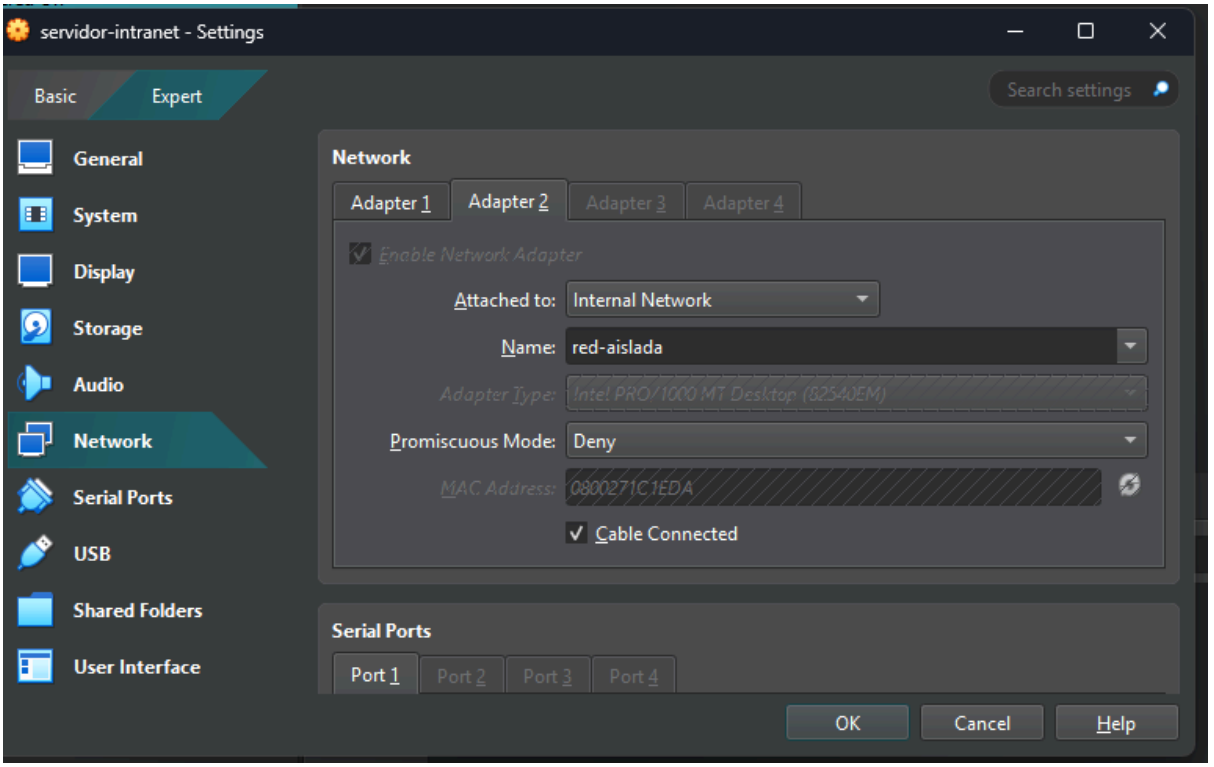


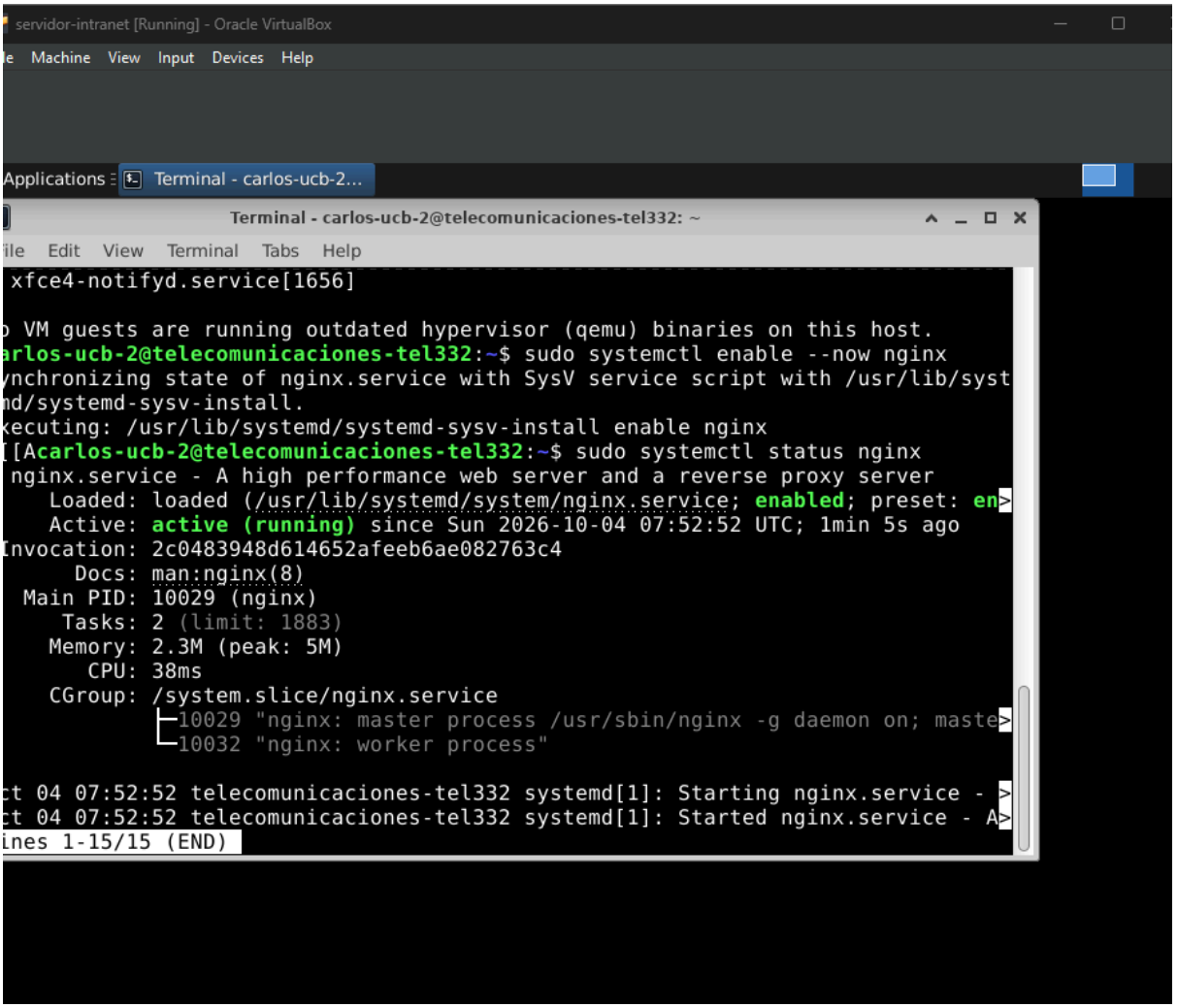
- **Instalé la segunda máquina**

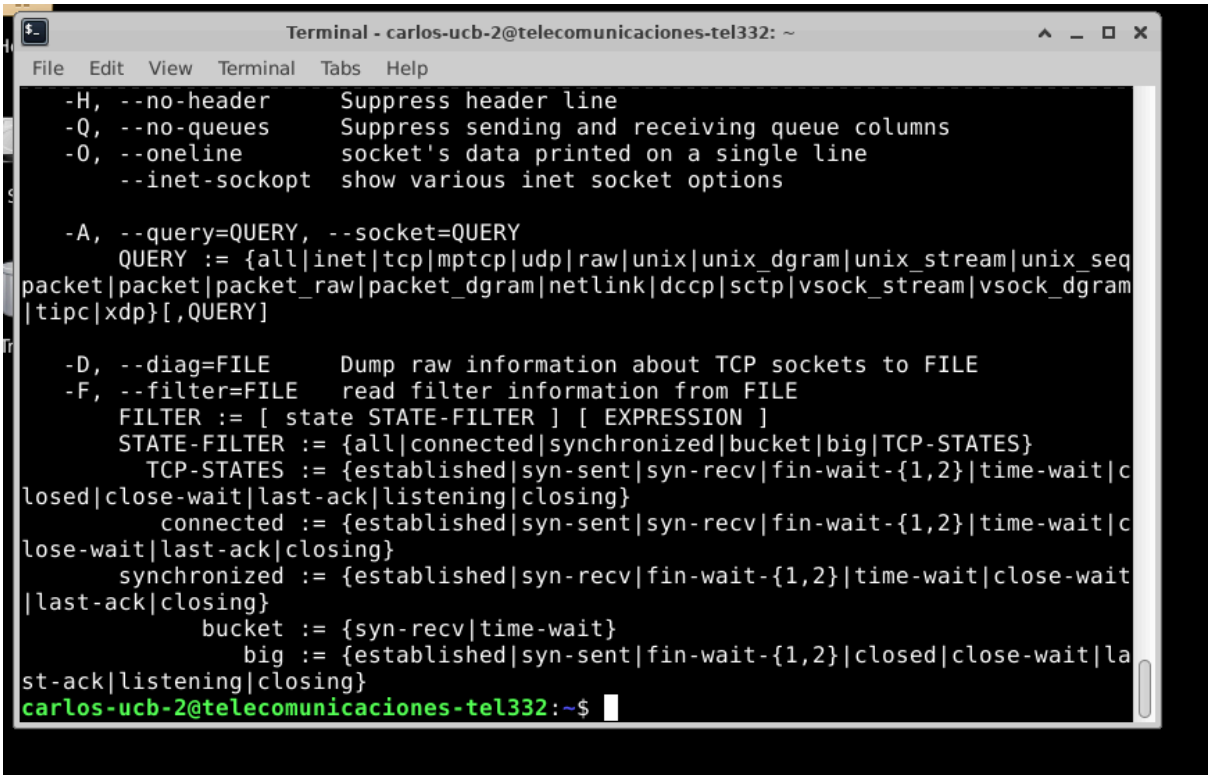


- **Las dos con red interna tel223-lab**

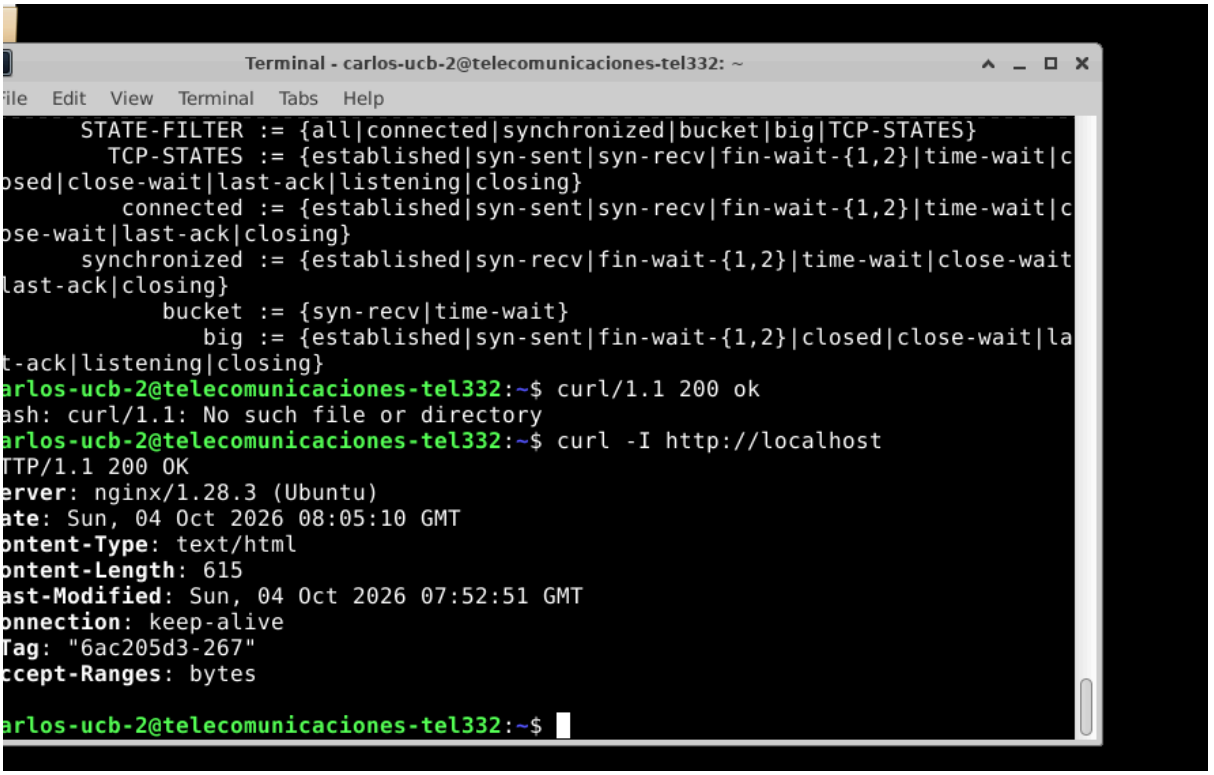

Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to:
Internal Network
Name:
red-aislada
Adapter Type:
Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode:
Deny
MAC Address:
0800270F9EFC
Cable Connected
Serial Ports
Port 1
Port 2
Port 3
Port 4
OK
Cancel
Help

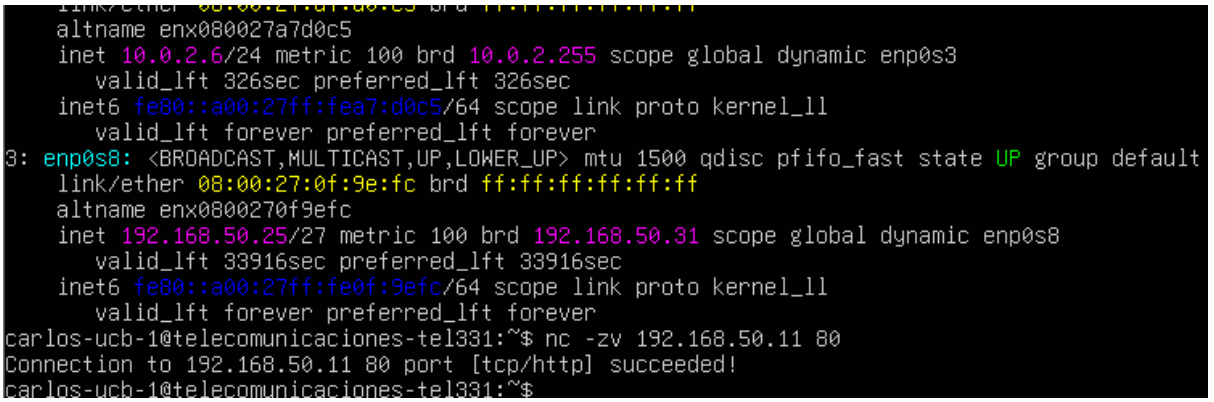

servidor-intranet - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to:
Internal Network
Name:
red-aislada
Adapter Type:
Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode:
Deny
MAC Address:
0800271C1EDA
Cable Connected
Serial Ports
Port 1
Port 2
Port 3
Port 4
OK
Cancel
Help


servidor-intranet [Running] - Oracle VirtualBox
Machine View Input Devices Help
Applications Terminal - carlos-ucb-2...
Terminal - carlos-ucb-2@telecomunicaciones-tel332: ~
File Edit View Terminal Tabs Help
xfce4-notifyd.service[1656]
VM guests are running outdated hypervisor (qemu) binaries on this host.
carlos-ucb-2@telecomunicaciones-tel332:~$ sudo systemctl enable --now nginx
Synchronizing state of nginx.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable nginx
[[Acarlos-ucb-2@telecomunicaciones-tel332:~$ sudo systemctl status nginx
nginx.service - A high performance web server and a reverse proxy server
Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: en
Active: active (running) since Sun 2026-10-04 07:52:52 UTC; 1min 5s ago
Invocation: 2c0483948d614652afeeb6ae082763c4
Docs: man:nginx(8)
Main PID: 10029 (nginx)
Tasks: 2 (limit: 1883)
Memory: 2.3M (peak: 5M)
CPU: 38ms
CGroup: /system.slice/nginx.service
10029 "nginx: master process /usr/sbin/nginx -g daemon on; maste
10032 "nginx: worker process"
Oct 04 07:52:52 telecomunicaciones-tel332 systemd[1]: Starting nginx.service -
Oct 04 07:52:52 telecomunicaciones-tel332 systemd[1]: Started nginx.service - A
lines 1-15/15 (END)

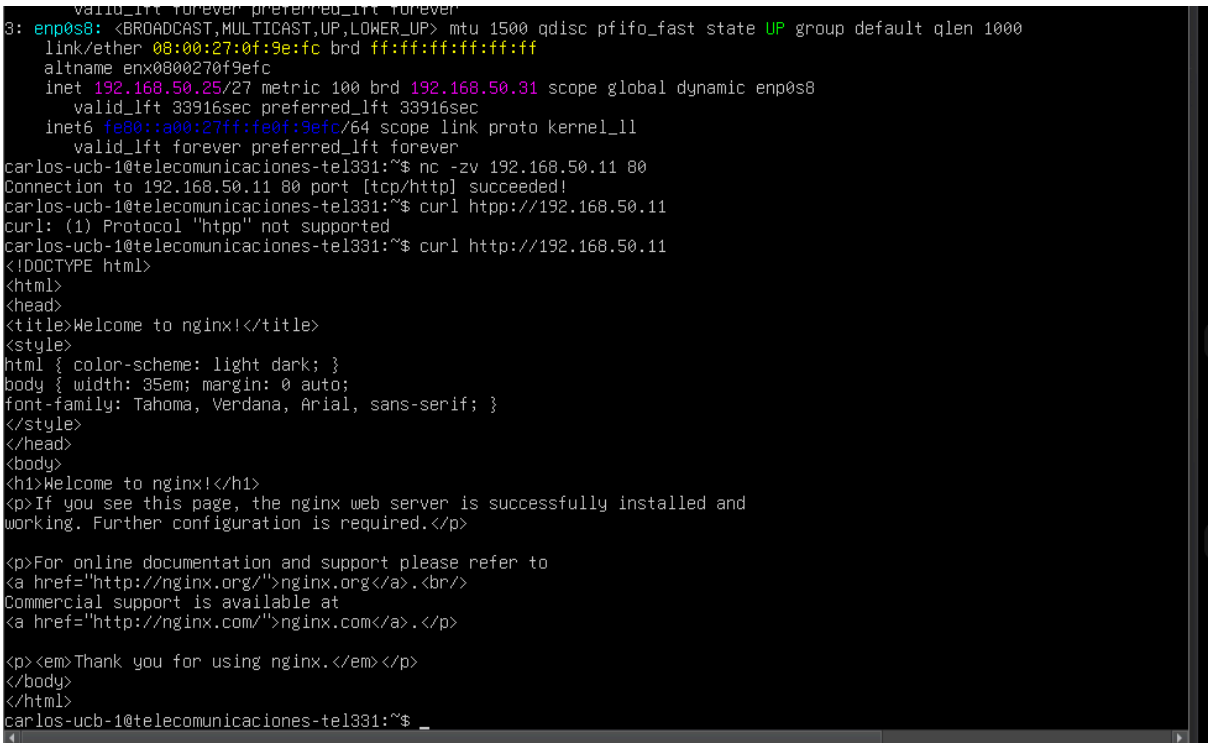

Terminal - carlos-ucb-2@telecomunicaciones-tel332: ~
File Edit View Terminal Tabs Help
-H, --no-header Suppress header line
-Q, --no-queues Suppress sending and receiving queue columns
-O, --oneline socket's data printed on a single line
--inet-sockopt show various inet socket options
-A, --query=QUERY, --socket=QUERY
QUERY := {all|inet|tcp|mptcp|udp|raw|unix|unix_dgram|unix_stream|unix_seqpacket|packet|packet_raw|packet_dgram|netlink|dccp|sctp|vsock_stream|vsock_dgram|tipc|xdp}[,QUERY]
-D, --diag=FILE Dump raw information about TCP sockets to FILE
-F, --filter=FILE read filter information from FILE
FILTER := [state STATE-FILTER] [EXPRESSION]
STATE-FILTER := {all|connected|synchronized|bucket|big|TCP-STATES}
TCP-STATES := {established|syn-sent|syn-recv|fin-wait-{1,2}|time-wait|closed|close-wait|last-ack|listening|closing}
connected := {established|syn-sent|syn-recv|fin-wait-{1,2}|time-wait|close-wait|last-ack|closing}
synchronized := {established|syn-recv|fin-wait-{1,2}|time-wait|close-wait|last-ack|closing}
bucket := {syn-recv|time-wait}
big := {established|syn-sent|fin-wait-{1,2}|closed|close-wait|last-ack|listening|closing}
carlos-ucb-2@telecomunicaciones-tel332:~$

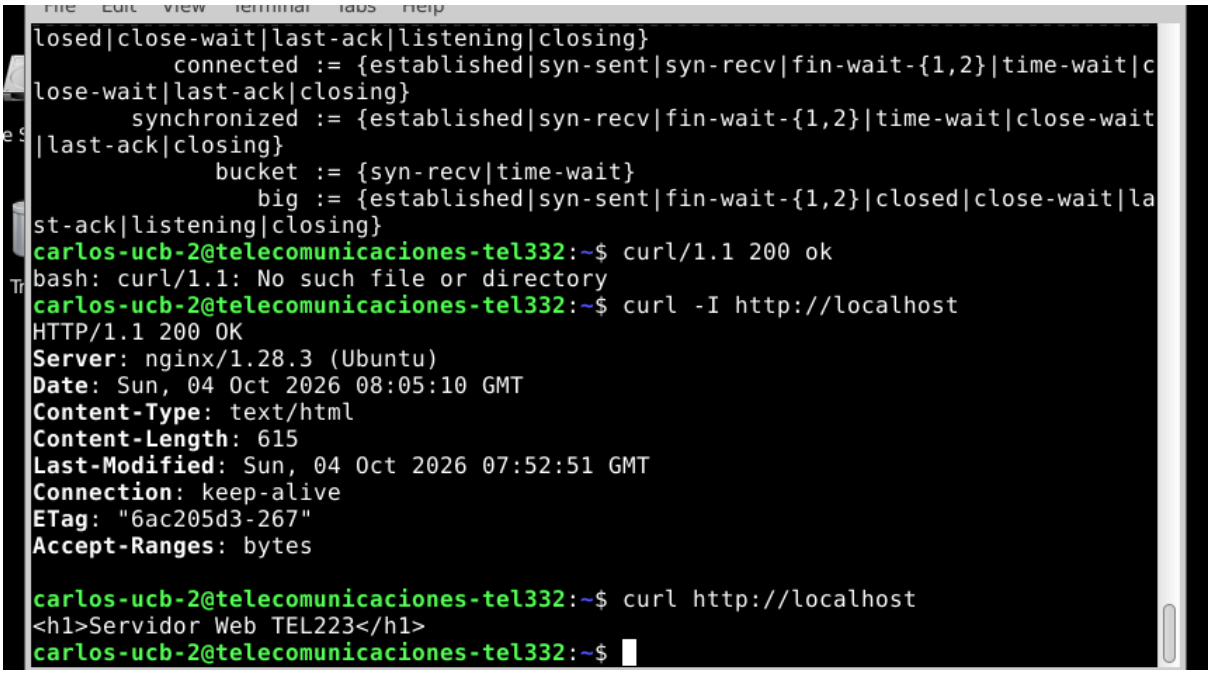
- **El ping cliente-servidor funcionó**



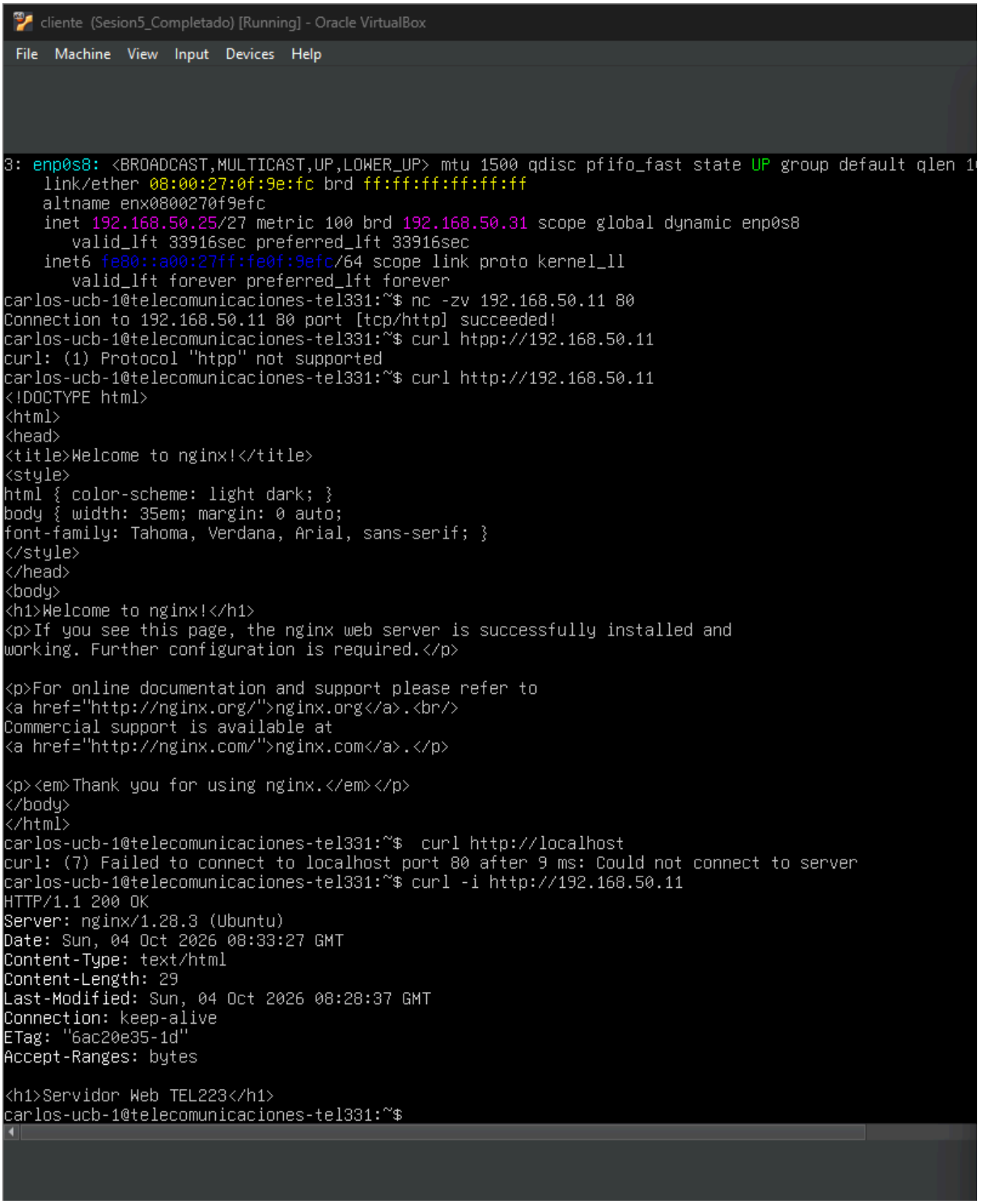


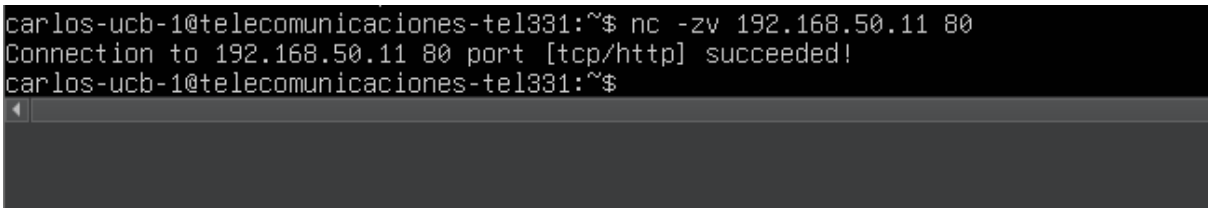
- **Publiqué una página con nginx**





- **Leí la transacción HTTP con curl**


cliente (Sesion5_Completado) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1
link/ether 08:00:27:0f:9e:fc brd ff:ff:ff:ff:ff:ff
altname enx0800270f9efc
inet 192.168.50.25/27 metric 100 brd 192.168.50.31 scope global dynamic enp0s8
valid_lft 33916sec preferred_lft 33916sec
inet6 fe80::a00:27ff:fe0f:9efc/64 scope link proto kernel_ll
valid_lft forever preferred_lft forever
carlos-ucb-1@telecomunicaciones-tel331:~$ nc -zv 192.168.50.11 80
Connection to 192.168.50.11 80 port [tcp/http] succeeded!
carlos-ucb-1@telecomunicaciones-tel331:~$ curl htpp://192.168.50.11
curl: (1) Protocol "htpp" not supported
carlos-ucb-1@telecomunicaciones-tel331:~$ curl http://192.168.50.11
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.
For online documentation and support please refer to
nginx.org.
Commercial support is available at
nginx.com.
Thank you for using nginx.
</body>
</html>
carlos-ucb-1@telecomunicaciones-tel331:~$ curl http://localhost
curl: (7) Failed to connect to localhost port 80 after 9 ms: Could not connect to server
carlos-ucb-1@telecomunicaciones-tel331:~$ curl -i http://192.168.50.11
HTTP/1.1 200 OK
Server: nginx/1.28.3 (Ubuntu)
Date: Sun, 04 Oct 2026 08:33:27 GMT
Content-Type: text/html
Content-Length: 29
Last-Modified: Sun, 04 Oct 2026 08:28:37 GMT
Connection: keep-alive
ETag: "6ac20e35-1d"
Accept-Ranges: bytes
<h1>Servidor Web TEL223</h1>
carlos-ucb-1@telecomunicaciones-tel331:~$


carlos-ucb-1@telecomunicaciones-tel331:~$ nc -zv 192.168.50.11 80
Connection to 192.168.50.11 80 port [tcp/http] succeeded!
carlos-ucb-1@telecomunicaciones-tel331:~$